IEEE Access
Multidisciplinary | Rapid Review | Open Access Journal

Draft of October 27, 2026 — not yet submitted.

Digital Object Identifier 10.1109/ACCESS.2026.DOI

VALDOMIRO DE O. SOUZA JR.[1], (Master's Degree Student), and RAFAEL KUNST[1]
[1]Applied Computing Graduate Program (PPGCA), Universidade do Vale do Rio dos Sinos (Unisinos), São Leopoldo 93022-750, Brazil (e-mail: valdomirojr@gmail.com, valdomiro.souza@gmail.com; rafaelkunst@unisinos.br)

Corresponding author: Valdomiro de O. Souza Jr. (e-mail: valdomirojr@gmail.com, valdomiro.souza@gmail.com).

This work was conducted within the Applied Computing Graduate Program (PPGCA) at Unisinos, Brazil.

Incident response (IR) in modern IT operations is constrained by two time-based indicators — Mean Time to Detect (MTTD) and Mean Time to Recovery (MTTR) — and by the cognitive load imposed on operations teams (SRE, NOC, SOC) by alert fatigue and telemetry complexity. Agentic AI systems, which combine Large Language Models (LLMs) with planning, memory, tool use, and human supervision, are increasingly proposed as *copilots* for these teams. This article reports a systematic literature review (SLR), conducted under the guidelines of Kitchenham et al. and the PRISMA 2020 flow, that consolidates the evidence on Agentic AI copilots for incident response, with cognitive-load reduction treated as a complementary outcome. Across two update cycles over a foundational corpus, 89 candidate studies (2020–2026) were identified and screened, 61 were assessed in full text by *three independent evaluators* (inter-rater agreement 90.2–91.8% per pair; Fleiss' $\kappa = 0.53$; 53/61 unanimous decisions), and **53 studies** were included. The review yields three main findings. First, no included study measures MTTD *and* MTTR nominally against its own baseline in production: the single study that reports nominal MTTR does so in a proof of concept, compared against the literature, with results unfavorable to the proposed agent. Second, the published evidence is dominated by detection proxies (accuracy, F1, AUC) and qualitative claims, while secondary sources report dramatic operational gains (e.g., MTTR reduced from 8 hours to 90 minutes) that remain unverified by primary experiments. Third, direct measurement of operator cognitive load exists in adjacent domains (neuroergonomics, control rooms, maritime operations) but has never been combined with an agentic copilot in IT incident response. These gaps jointly define a concrete research agenda: field evaluations of agentic copilots instrumented for nominal MTTD/MTTR with their own baselines and validated cognitive-load instruments.

Agentic AI, AIOps, cognitive load, incident response, LLM agents, multi-agent systems, MTTD, MTTR, SOC, systematic literature review.

MODERN IT operations teams face a structural asymmetry: distributed, cloud-native systems generate telemetry at machine speed, while detection, diagnosis, and remediation remain bounded by human cognition. The operational cost of this asymmetry is captured by two indicators — *Mean Time to Detect* (MTTD) and *Mean Time to Recovery* (MTTR) — and by a human cost that is harder to see on dashboards: alert fatigue and sustained cognitive load on SRE, NOC, and SOC personnel. Industry reports cited in the corpus place average breach identification and containment at 277 days (204 to detect, 73 to contain) [1], while SOC analysts discard the majority of alerts as false positives.

Agentic AI — AI systems that pursue goals with some degree of autonomy by combining LLM-based reasoning with planning, memory, tool use, and human oversight [2], [3] — has been proposed as a new operating model for incident response: not full automation, but a *copilot* that perceives, reasons, acts, and learns alongside the human operator. The promise is threefold: earlier detection (MTTD), faster remediation (MTTR), and a lighter day-to-day workload for operations teams.

This article asks whether the published evidence supports that promise. It reports a systematic literature review (SLR)

conducted under the guidelines of Kitchenham et al. [4], with study selection reported in the PRISMA 2020 format [5], scope delimited with the PICOC framework [6], and quality assessment mirroring the DARE criteria [7]. The review was executed as an *update review* in two cycles over a foundational corpus of 19 studies, processing 89 candidate studies in total, and — unusually for SLRs in this area — every full-text assessment was performed independently by *three evaluators*, with inter-rater reliability quantified by Cohen's [8] and Fleiss' [9] kappa.

The contributions are:

1) a consolidated corpus of 53 studies (2020–2026) on Agentic AI for incident response and adjacent IT operations, organized into a taxonomy of agentic paradigms (Section IV);
2) triple-verified bibliometrics (OpenAlex, Crossref, Scopus) and an internal cross-citation map of the corpus (52 citing edges) identifying its conceptual hubs;
3) evidence synthesis showing that **no study measures MTTD and MTTR nominally with its own baseline in production** (Section IV-C), and a catalog of the proxies actually used;
4) a bridge to the human-factors literature: a cluster of studies that measure operator cognitive load directly — yet never in combination with an agentic copilot in IT incident response (Section IV-D);
5) a methodological contribution: a fully reproducible three-rater protocol whose residual disagreements (8 of 61 decisions) are all traceable to three explicit protocol rules rather than to content interpretation (Section IV-F).

A. FROM AIOPS TO AGENTIC AI

AIOps applies machine learning to IT operations data — logs, metrics, traces — for anomaly detection, root cause analysis (RCA), and remediation. Surveys in the corpus trace its evolution from knowledge-based automation to ML pipelines [10] and, after 2023, to LLM-centered architectures [11]. The agentic turn adds goal pursuit under governance: taxonomies distinguish *AI agents* (tool-using, single-task) from *Agentic AI* (multi-step autonomy with planning and memory) [2], and reference surveys map its technologies, applications, and societal implications [3], [12], [13].

B. INCIDENT RESPONSE AND ITS METRICS

Incident response spans detection, triage, diagnosis (RCA), remediation, and post-incident learning. MTTD and MTTR are its canonical outcome metrics; alert fatigue and operator cognitive load are its canonical human costs. In security operations, the same cycle appears as SOC triage and SOAR playbooks [14], [1]. A persistent finding of this review — anticipated here — is that primary studies rarely instrument these nominal metrics, reporting detection-quality proxies instead.

PICOC delimitation of the review.

Element	Definition
Population	Complex software systems and the teams operating them (SRE, NOC, SOC); microservices; IT operations pipelines.
Intervention	Agentic AI as copilot or autonomous agent: LLM-based agents, multi-agent systems, LLM4AIOps, the perceive–reason–act–learn loop.
Comparison	Traditional automation, non-agentic AIOps, or human baselines — *declared only with an executed empirical baseline* (rule v1.1.0).
Outcomes	MTTD, MTTR, observability, detection accuracy, alert fatigue, and operator cognitive load (complementary outcome).
Context	The incident-response life cycle in IT operations and cybersecurity; HITL/HOTL supervision; production vs. simulation.

C. THE FOUNDATIONAL CORPUS AND ITS HUBS

The present review updates a foundational SLR (19 studies, P01–P19). Cross-citation analysis over the full corpus (Section IV-E) shows that three foundational surveys act as conceptual hubs — [3] (7 internal citations), [2] (6), and [15] (5) — confirming their role as the definitional substrate on which the empirical studies build.

The protocol follows Kitchenham et al. [4]: planning (PICOC, research questions, search string), conducting (search, screening, quality assessment, extraction), and reporting (this article). All protocol artifacts — prompts, extraction sheets, per-study assessments, and machine-readable result tables — are version-controlled in the research repository.

A. PICOC

Table 1 shows the PICOC delimitation [6]. A protocol rule (v1.1.0), introduced after a three-rater calibration of the PICOC extraction itself, requires that *Comparison* be recorded as declared **only when the study executes an empirical comparison against a concrete baseline**; purely conceptual contrasts (e.g., ''Agentic AI vs. traditional AI'' in prose) are recorded as not applicable. This rule eliminated 100% of the inter-rater divergences observed in the Comparison field.

B. RESEARCH QUESTIONS

- **RQ1 (Context/Definitions).** Which capabilities and autonomy levels (planning, memory, tool use, human supervision) characterize Agentic AI, and how do they affect decision-making in complex environments?
- **RQ2 (Engineering/Architecture).** Which architectures and mechanisms (orchestration, memory, tools, guardrails, observability) sustain Agentic AI in production?
- **RQ3 (Evidence/Benefits).** Which quantitative and qualitative evidence reports benefits, and which metrics are

used (response time, decision quality, operational load), especially in incident response?

- **RQ4 (Challenges/Ethics).** Which risks and challenges (security, robustness, governance, accountability) limit responsible use in critical systems?
- **RQ5 (Gaps).** Which gaps and future directions (evaluation/benchmarking, threat models, governance, observability, alignment) are identified?

C. SEARCH AND IDENTIFICATION

The search string was derived from the cross-study PICOC synthesis as two mandatory blocks — *Intervention* (agentic AI, LLM agents, multi-agent systems, AIOps copilots) AND *Context/Population* (incident response, IT operations, SRE/NOC/SOC, observability) — with *Outcomes* deliberately optional, since only 1 of 38 calibration studies named MTTD/MTTR in its metadata. The string was calibrated against the known corpus and validated by real execution on OpenAlex (≈49,700 records) and on the Scopus Search API (12,783 records, 2020+, English), achieving 13/14 recall of the included calibration set in both engines with the same single miss. Candidate identification combined assisted search (cycle 1: three independent LLM assistants, ≈51 raw records) and author-driven discovery (cycle 2: 49 candidates). Backward/forward snowballing is mandatory in the protocol because one included study is not indexed by Scopus and one is not retrieved by the main query.

D. SELECTION CRITERIA

Inclusion required: publication year ≥2020 (I1); an identifiable peer-reviewed venue (I2); venue quality thresholds (I3–I4, national Qualis A1–A2 and SJR Q1–Q2); at least one citation in any of three indexers, with a recency exception for papers younger than 12 months (I5); and thematic adherence — an *agentic* approach in the *IR/AIOps/SOC* domain (I6). Exclusion criteria: sub-threshold venue strata (E1), non-agentic approaches (E2), out-of-domain studies (E3), duplicates (E4), and non-peer-reviewed material (E5). A dated protocol amendment in cycle 2 allowed four candidates outside the I3/I4 strata to proceed to full-text assessment *with a mandatory stratum caveat*; the amendment is itself part of the audit trail and, notably, is the root cause of three of the eight residual inter-rater divergences (Section IV-F).

E. RATIFIED PROTOCOL RULES (R1–R3)

The inter-rater analysis (Section IV-F) traced every residual evaluator divergence to three recurring boundary questions. After that analysis, the review conductor ratified the following rules, which are consistent with — and now normatively bind — the published corpus decisions:

- **R1 (non-agentic secondary studies).** Secondary studies (surveys/SLRs) whose object is *not* Agentic AI are excluded from the primary corpus regardless of methodological quality; agentic-centered secondary studies are retained as *foundational-conditional* background (e.g., [16], [17], [18], [13], [1]). This ratifies the exclusion of the non-agentic RCA and log-anomaly reviews assessed in cycle 1.
- **R2 (normative placement of amendments).** Eligibility amendments bind only when written into the *normative Stage-1 text* of the assessment protocol, not into its input metadata. The cycle-2 stratum amendment is ratified under this rule: the four sub-stratum candidates keep their full assessments with a mandatory stratum caveat attached to any use of their evidence.
- **R3 (domain boundary and the human-factors cluster).** Out-of-domain studies are excluded from the primary corpus even when methodologically strong; the human-factors cluster [19], [20], [21], [22] is retained strictly as *complementary evidence for the cognitive-load outcome* (Table 4) and never feeds the agentic research questions (RQ1, RQ2, RQ4); a cluster study with no RQ adherence at all remains excluded.

Selection flow across the two update cycles.

Stage	Cycle 1	Cycle 2
Identified	≈51 → 21 unique	49
Screened	20 (–1 duplicate)	49
Excluded at screening	2 (E1)	27 (E2=19, E3=8)
Assessed in full text	18	22
Excluded after assessment	4	2
Included	**14**	**20**
Corpus (cumulative)	33 (19+14)	**53** (33+20)

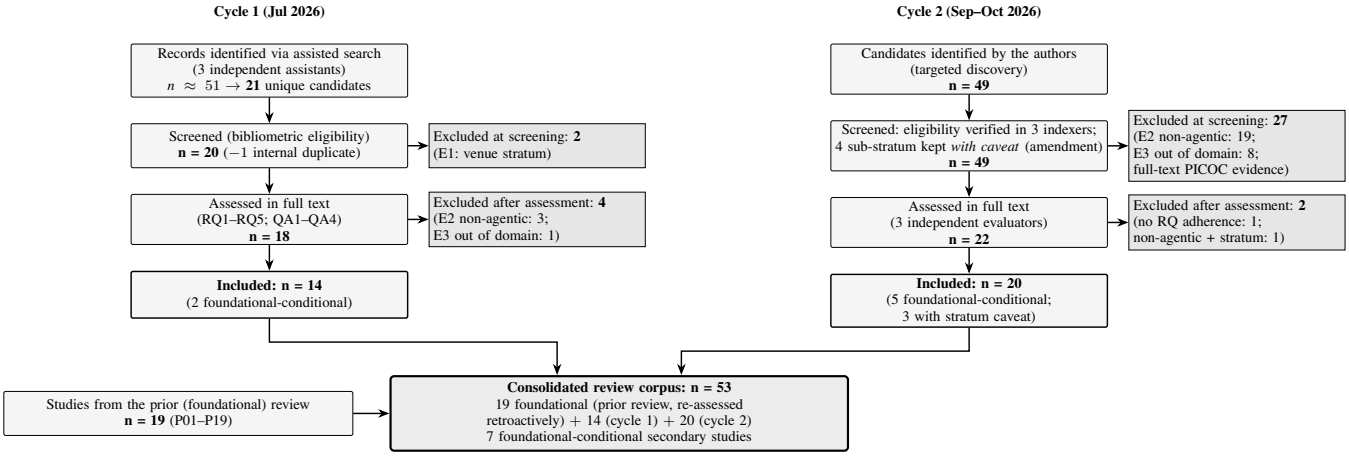
F. SELECTION FLOW (PRISMA)

Fig. 1 shows the PRISMA 2020 flow of both update cycles over the foundational corpus, and Table 2 summarizes the stage counts side by side. In cycle 2, thematic exclusion (E2/E3) was applied as early as the screening stage, because a full-text PICOC extraction of all 49 candidates (one reading agent per PDF, consolidated and audited) provided per-study evidence before assessment — each screening exclusion cites that evidence and a cycle-1 precedent.

The 53-study corpus comprises the 19 foundational studies, 14 cycle-1 inclusions, and 20 cycle-2 inclusions; seven of these are secondary ''foundational-conditional'' studies [16], [17], [18], [13], [23], [24], [1] that migrate to background material if the protocol is restricted to primary studies (primary corpus: 46).

G. QUALITY ASSESSMENT AND EXTRACTION

Each assessed study received: (i) a verified bibliometric record (DOI, venue, year, and citation counts kept *per source* — OpenAlex, Crossref, Scopus — never summed, with ''not indexed'' distinguished from zero); (ii) an RQ-coverage score (RQ1–RQ5, each rated fully/partially/not answered = 1.0/0.5/0.0, max 5.0); (iii) a DARE-style quality score (QA1 clear objectives, QA2 replicable methodology, QA3 solid evidence base, QA4 coherent conclusions; max 4.0; bands High ≥3.0, Medium 1.5–2.5, Low <1.5); and (iv) an 11-field



PRISMA 2020 flow of the two update cycles over the foundational corpus (adapted for a review update). Cycle-2 screening exclusions are anchored in a full-text PICOC extraction of all 49 candidates; the stratum amendment is dated and auditable in the protocol.

structured extraction sheet with per-claim evidence anchors (section/page). The 19 foundational studies were re-assessed retroactively under the same rubric — without re-opening their inherited inclusion — so that all 61 assessments are directly comparable.

H. THREE-EVALUATOR PROTOCOL

Every full-text assessment was executed independently by three LLM-based evaluators (Claude, ChatGPT, Gemini) from the same written prompt and PDF. The agreement set comprises 61 decisions: the 19 foundational re-assessments, all 20 screened cycle-1 candidates (the two stratum-ineligible candidates also received full assessments, recorded as exclusions), and the 22 cycle-2 assessments. Agreement was computed on collapsed include/exclude decisions with Cohen's κ per pair [8] and Fleiss' κ overall [9]. Results are reported in Section IV-F.

A. CORPUS OVERVIEW AND TAXONOMY

Table 3 organizes the 53 included studies by agentic paradigm. Three observations structure the corpus. First, the *foundational layer* is dominated by definitional surveys [2], [3], [12], [13] and governance/trust reviews [25], [26]; these answer RQ1/RQ4/RQ5 strongly but contribute no empirical benefit evidence. Second, the *empirical core* is concentrated in cycle 1: multi-agent LLM systems for RCA and SOC [27], [14], [28], [29], closed-loop remediation [30], and a small-language-model agent for ICT operations [31] — the methodological high-water marks of the corpus (QA 3.5–4.0 of 4.0). Third, cycle 2 adds the copilot pattern proper — RAG-based security copilots [32], [33], [34] and SOC hyperautomation [35] — plus a *human-factors cluster* absent from earlier cycles [19], [20], [21], [22].

B. RQ COVERAGE AND QUALITY

Fig. 2 shows per-RQ coverage across the three study sets. RQ coverage shifted between cycles in a telling way. In

Taxonomy of the included corpus (n=53; representative studies).

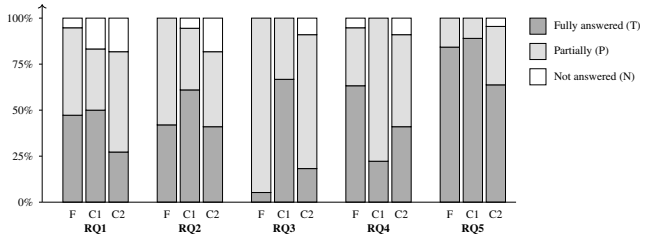
Paradigm	Studies
LLM multi-agent (RCA/SOC)	[27], [14], [28], [29], [35]
Single-agent LLM copilot (+RAG)	[36], [32], [33], [34]
Closed-loop remediation agents	[37], [30], [38]
SLM/compact-model agents	[31]
MAS/RL and self-organization (non-LLM)	[39], [40]
Conceptual agentic frameworks (AIOps/PdM)	[41], [42]
AI-assisted SIEM / alert triage	[43]
Foundational surveys (agents, LLM-sec, AIOps)	[11], [2], [3], [15], [12], [10], [16], [17], [18], [13], [23], [24], [1]
Human factors & instruments	[44], [19], [20], [21], [22]

cycle 1 (18 assessed; mean RQ score 3.83, mean QA 3.50) the systematic gap was **RQ4** (ethics and governance: only 4/18 fully answered). In cycle 2 (22 assessed; mean RQ 3.16, mean QA 2.73; zero unconditional inclusions) the gap moved to **RQ3/QA3**: abundant conceptual contribution with scarce empirical benefit evidence. The retroactive assessment of the foundational studies (mean RQ 3.66, mean QA 2.53) shows the same profile — high RQ coverage by surveys, low QA for lack of their own validation. Across all 61 assessments the pattern is stable: *the literature is conceptually rich and empirically thin precisely on the benefit claims that motivate it*.

C. EVIDENCE ON MTTD AND MTTR

This is the review's central result. After screening 89 candidates and assessing 61 in full text across two cycles:

- **No study measures MTTD and MTTR nominally against its own baseline in production.**
- **One study reports nominal MTTR.** The SOC hyperautomation agent of [35] reports per-phase remediation times of ≈1–1.5 min in a proof of concept (one use case, SSH brute force), compared against *literature values*



Coverage of the research questions by set (share of assessed studies rating each RQ fully/partially/not answered; F = foundational, n=19; C1 = cycle 1, n=18; C2 = cycle 2, n=22). The empirical core of cycle 1 is the only set in which RQ3 (evidence of benefits) is predominantly fully answered (12/18); the foundational surveys answer RQ3 fully in just 1/19 cases, and cycle 2 in 4/22 — while RQ4 (ethics and governance) is the systematic gap of cycle 1 (4/18) and RQ5 (gaps and directions) is strong everywhere.

for a commercial security copilot (≈30–45 s) — i.e., no baseline of its own, and a result *unfavorable* to the proposed agent. The authors' candor is valuable; the evidence remains at proof-of-concept level.

- **Secondary evidence reports dramatic gains.** The SOC survey [1] relays primary reports of MTTR reduced from 8 hours to 90 minutes, 98% of alerts auto-triaged with containment under 5 minutes, and a 277-day industry detect-and-contain baseline. None of these numbers is reproduced by a primary study in the corpus.
- **Proxies dominate.** The strongest quantified operational outcomes are proxies: a 74–84% global alert-volume reduction in a national SOC [43]; operator reaction time falling from ≈276 s to ≈107 s with decision support in a control-room experiment (Wilcoxon $p < 0.01$) [22]; end-to-end incident-report latency of 2.9–11.6 s with 0.3 s pre-detection [34]; and copilot answer quality (BERTScore ≈85–87%) without timing [33]. Detection studies report accuracy/F1/AUC only.

The gap is therefore not merely a matter of "few studies": it is a structural mismatch between the motivating claim of the field (time-metric reduction) and its measurement practice (detection proxies and qualitative assertions). The claim that agentic copilots reduce MTTD and MTTR is, as of 2026, *plausible but unverified by primary evidence*.

D. COGNITIVE LOAD: THE COMPLEMENTARY OUTCOME

The second gap concerns an outcome this review treats as complementary rather than incidental: operator cognitive load. The motivating observation was already present in the foundational review — the explosion of observability data (metrics, logs, traces) has raised the cognitive load of operations teams faster than tooling has relieved it [11], [10] — yet, within IT incident response, *no corpus study measures cognitive load directly*; alert fatigue appears only as an alert-volume proxy [43]. Direct measurement does exist in adjacent domains, and cycle 2 deliberately retained this human-factors cluster (with stratum caveats under the protocol amendment): neuroergonomic foundations of mental workload and engagement [19]; a multimodal workload database with classification AUC up to 0.998 and NASA-TLX ground truth [21]; and a human experiment in which an AI decision-support system improved control-room operator performance [22]. The corpus also offers a validated usability instrument for conversational agents (BUS-15) [20].

A cognitive-load measurement toolbox for agentic-copilot studies, assembled from the corpus.

Instrument	Corpus evidence	Recommended use with a copilot
NASA-TLX (subjective workload)	Ground truth for multimodal workload classification [21]	Pre/post per incident shift, copilot on vs. off
Physiological multimodal (EEG, ECG, breathing)	Classification AUC up to 0.998 under activity [21]; neuroergonomic framework [19]	Continuous workload index during on-call duty; validate against NASA-TLX
Reaction / response time	Operator reaction time ≈276→107 s with AI decision support (Wilcoxon $p < 0.01$) [22]	Time-to-first-action per alert as the operator-level analogue of MTTD
Alert-volume exposure	74–84% global alert reduction in a national SOC [43]	Alerts per analyst-hour as the fatigue *exposure* covariate
Conversational usability (BUS-15)	Validated 15-item scale for AI-based conversational agents [20]	Perceived quality of the copilot interface; moderator of workload effects

Table 4 consolidates these assets into a *measurement toolbox* for future copilot studies — in our view, the most reusable contribution this corpus makes to the cognitive-load question. The decisive observation is that the intersection *agentic copilot* × *IT incident response* × *measured cognitive load* is empty: every instrument in Table 4 has been validated, but never in the presence of an agentic copilot in IT operations. Together with Section IV-C, this defines the experimental agenda that motivates the authors' ongoing work: a copilot evaluated in the field with nominal MTTD/MTTR against its own baseline *and* the instruments of Table 4.

E. CROSS-CITATION STRUCTURE OF THE CORPUS

Internal citation analysis (triple-checked: OpenAlex `referenced_w` Crossref reference lists, Scopus `REF()` by title; union of 52 citing edges) shows a coherent field rather than a disconnected collection of papers. The hubs are the definitional surveys — [3] (7 internal citations), [2] (6), [15] (5) — followed by the AIOps/MLOps and log-anomaly surveys [10], [45] (3 each). The most connected citer is the multi-agent SOC system of [14] (7 outgoing edges, five of them to cycle-2 candidates), and cycle-2 studies cite back into the corpus ([1]→[37], [32]; [35]→[46], [33]), indicating that the two cycles sample one evolving conversation. Source-specific quirks are material to reproducibility: Scopus resolves references by *title*, not DOI; one open-access venue is not indexed there at all; and preprint DOIs can split citation identity across

Inter-rater agreement on include/exclude decisions (n=61).

	A×B	A×C	B×C
Agreement	90.2%	91.8%	91.8%
Cohen's κ	0.57	0.51	0.51

Fleiss' κ (three raters): **0.53**; unanimous decisions: **53/61**.

sources (handled here by alias resolution). Fig. 3 renders the full network.

F. INTER-RATER RELIABILITY

Table 5 reports agreement over the 61 collapsed include/exclude decisions. Pairwise agreement is 90.2–91.8% (κ 0.51–0.57); Fleiss' $\kappa = 0.53$; **53/61 decisions are unanimous**, including all 19 foundational re-assessments and three unanimous exclusions. The κ values are deflated by prevalence (89.6% of all verdicts are "include"), so the substantive reading is the nature of the eight divergences: *all eight are protocol-rule divergences, none is a content divergence* — in every case the three evaluators' scores agree and only the consequence rule differs. They fall into exactly three families: (A) the treatment of non-agentic secondary studies (2 cases); (B) whether the cycle-2 stratum amendment binds, which one evaluator — applying the strict criterion — did not honor (3 cases); and (C) domain/stratum boundary calls (3 cases). These three families were subsequently ratified as protocol rules R1–R3 (Section III-E); re-reading the 61 decisions under the ratified rules eliminates all residual divergence. Mean absolute score differences are small (RQ 0.44–0.63; QA 0.31–0.63), with systematic evaluator profiles: one strict on protocol and quality, one lenient, one intermediate with the hardest thematic cut.

A. WHAT THE EVIDENCE SUPPORTS — AND WHAT IT DOES NOT

The corpus firmly supports three claims: (i) a consolidated conceptual model of Agentic AI for operations (autonomy levels, orchestration, guardrails, HITL) [2], [3], [12], [13]; (ii) technical feasibility of agentic RCA, triage, and remediation at prototype level, with strong detection-quality results [31], [30], [27], [14], [28], [29]; and (iii) a well-articulated risk and governance agenda [25], [47], [26], [15]. It does *not* yet support the headline operational claim: the MTTD/MTTR reduction attributed to agentic copilots rests on secondary reports and qualitative assertions (Section IV-C). Reviewers and practitioners should treat vendor-scale numbers (e.g., 8h → 90min MTTR) as hypotheses to be tested, not as established effects.

B. WHY THE MEASUREMENT GAP PERSISTS

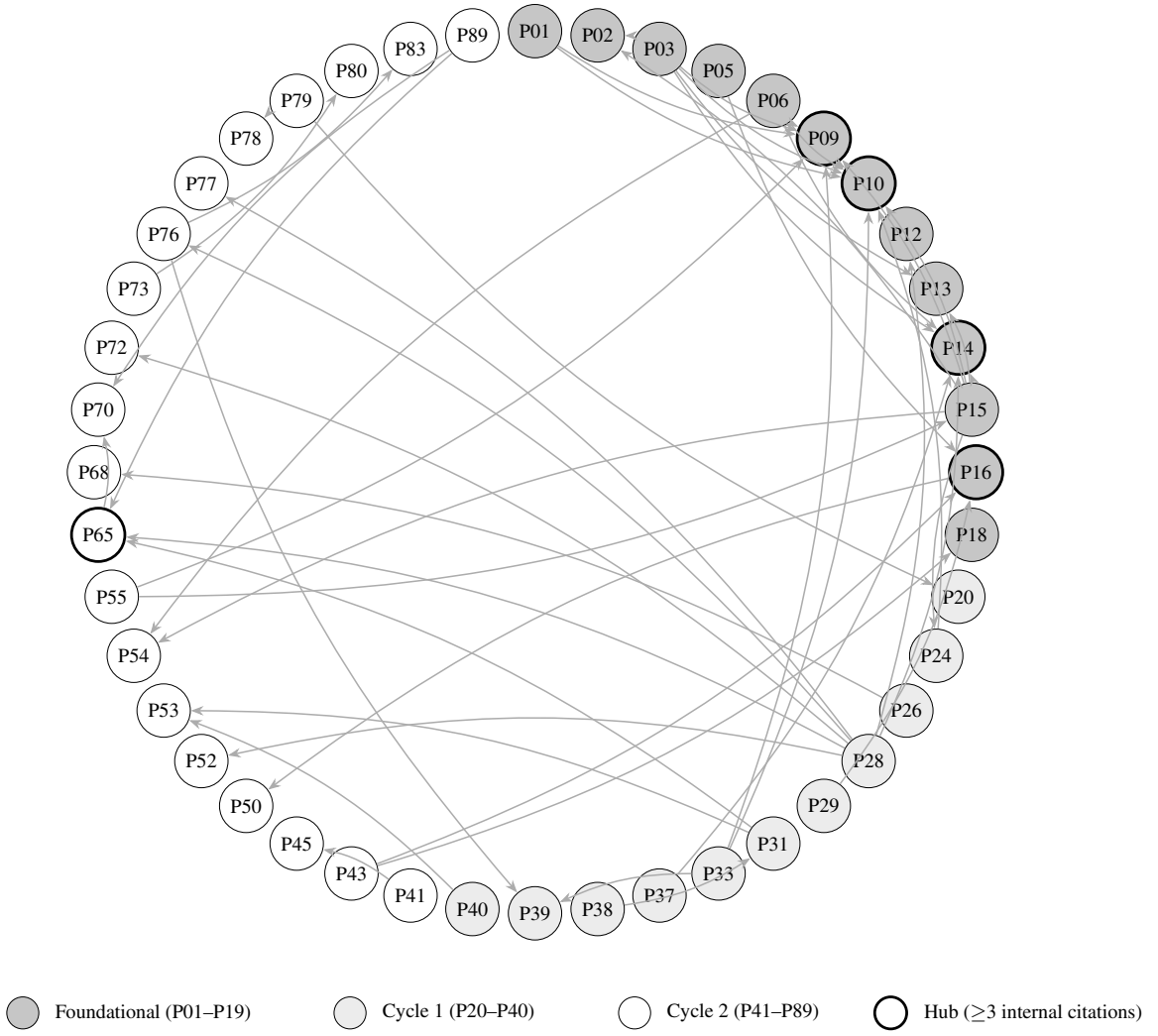
Three causes are visible in the corpus: production access (most studies evaluate on public datasets or testbeds; industrial deployments are rare and rarely instrumented for time metrics); benchmark culture (detection communities optimize F1/AUC, which do not compose into end-to-end time outcomes); and the autonomy–accountability tension (studies that do deploy agents constrain them to low-risk actions, making time gains modest and hard to attribute). That the only available nominal measurement [35] is *unfavorable* to the agent is a useful counterweight to publication bias.

C. RESEARCH GAPS CONSOLIDATED ACROSS BOTH REVIEWS

The foundational review synthesized six gap axes for agentic IR; the two update cycles allow each to be re-examined against newer evidence. **(i) From recommendation to safe autonomous execution.** Remediation in LLM4AIOps was found to stop at recommendation and triage [11], [12]; the update cycles confirm the axis — closed-loop execution exists at prototype level [30], [38] and at proof-of-concept level in SOC hyper-automation [35], but verifiable, auditable, reversible execution in production remains unreported. **(ii) Complex telemetry (traces).** The difficulty of feeding distributed traces to LLMs [11] persists: cycle-2 primary studies operate on logs, alerts, and flows [48], [43], [34]; trace-native agentic RCA remains open. **(iii) Memory integrity and contradictory knowledge.** Memory poisoning and conflicting sources were flagged as structural risks [25]; no update-cycle study evaluates agent-memory integrity empirically — the axis is untouched. **(iv) Governance, accountability, and adversarial risk.** The richest axis in volume: governance frameworks and threat taxonomies [25], [47], [15], [26], [23], [24] and HITL adoption factors [49], [44], yet still without quantitative evaluation of governance mechanisms under incident conditions. **(v) Causal reasoning vs. statistical correlation.** Agentic RCA advanced with graph- and Bayesian-structured approaches [38], [29], [48], partially addressing the axis, though evaluation remains on curated failure sets. **(vi) Resource efficiency and sustainability.** The SLM agent of [31] is still the corpus's only direct response (compact models, confidentiality, cost); the axis otherwise remains a statement of intent. Axes (i), (iii), and (vi) are therefore the least served by three years of literature — a concrete map for future primary studies, to which this review adds a seventh axis: *measured cognitive-load outcomes* (Section IV-D).

D. IMPLICATIONS FOR RESEARCH AND PRACTICE

For researchers, the agenda is concrete: field studies that instrument MTTD and MTTR nominally, pre/post or A/B against the team's own baseline, with the agentic copilot as the treatment; and that pair time metrics with validated human-factors instruments (NASA-TLX [21], BUS-15 [20]) to test the cognitive-load "bonus" directly. For practitioners, the corpus supports piloting copilots on triage and report drafting (where latency and quality evidence exists [33], [34]) under HITL governance [49], [44], while treating full-loop autonomous remediation as experimental [30], [35].



Internal citation network of the corpus (52 edges, A→B = "A cites B"), triple-checked across OpenAlex, Crossref, and Scopus. Definitional surveys [3], [2], [15] are the main hubs; the multi-agent SOC system [14] is the most connected citer (7 outgoing edges, five of them to cycle-2 studies), and cycle-2 studies cite back into the corpus — evidence of one evolving research conversation rather than disjoint samples.

E. METHODOLOGICAL LESSONS FOR LLM-ASSISTED SLRS

Two findings generalize beyond this topic. First, three independent LLM evaluators under one written protocol achieve high percentage agreement (90.2–91.8% pairwise) with moderate chance-corrected κ, and their residual disagreements expose *protocol ambiguities, not model randomness* — a cheap and effective audit mechanism. Second, protocol amendments must be placed in the normative body of the evaluation prompt, not in its input metadata: the one evaluator that missed the stratum amendment read it as advisory, producing three of the eight divergences.

Selection. Assisted identification (LLM search in cycle 1, author-driven discovery in cycle 2) may bias candidate coverage; this is mitigated by dual-engine string validation (13/14 recall on OpenAlex and Scopus), mandatory snowballing, and full audit trails, but a database-exhaustive census was not the goal of the update design. **Strata criteria.** Venue-quality thresholds (I3–I4) use a national classification (Qualis) alongside SJR; the dated amendment that retained four sub-threshold studies is explicit, and all conclusions in Sections IV-C–IV-D hold with those studies removed. **Evaluator validity.** All three raters are LLM systems; they are independent of each other but share potential model-family biases. The unanimity of the core corpus (87%), the protocol-only nature of divergences, and per-claim evidence anchors in every assessment mitigate — but do not eliminate — this threat; the three protocol questions exposed by the divergences were adjudicated by the (human) review conductor and ratified as rules R1–R3 (Section III-E). **Bibliometric sources.** Citation counts are kept per source and never summed; source quirks (title-based reference resolution, unindexed venues, preprint DOI splits, API service-level limits) are documented in the audit trail.

This SLR consolidated 53 studies (2020–2026) on Agentic AI as a copilot for incident response, assessed by three independent evaluators with high agreement (53 of 61 decisions unanimous). The field has matured conceptually — shared definitions, reference architectures, governance agendas, and a coherent citation structure — and demonstrates prototype-level feasibility of agentic detection, RCA, and remediation. Yet its motivating claim remains unmeasured: no primary study reports nominal MTTD and MTTR against its own production baseline, and operator cognitive load has never been measured in the presence of an agentic copilot in IT operations. The most valuable next contribution to this literature is therefore not another architecture, but a field evaluation: an agentic copilot deployed with a team, instrumented for MTTD/MTTR with its own baseline, and paired with validated workload instruments. That study is the direct continuation of this review.

The authors thank the Applied Computing Graduate Program (PPGCA) at Unisinos. Review artifacts (protocol, prompts, assessments, extraction sheets, and machine-readable tables) are maintained in a version-controlled repository and are available from the authors upon reasonable request.

[1] S. Srinivas, B. Kirk, J. Zendejas, M. Espino, M. Boskovich, A. Bari, K. Dajani, and N. Alzahrani, "AI-Augmented SOC: A Survey of LLMs and Agents for Security Automation," *Journal of Cybersecurity and Privacy*, 2025.
[2] R. Sapkota, K. I. Roumeliotis, and M. Karkee, "AI Agents vs. Agentic AI: A Conceptual taxonomy, applications and challenges," *Information Fusion*, 2025.
[3] D. B. Acharya, K. Kuppan, and B. Divya, "Agentic AI: Autonomous Intelligence for Complex Goals — A Comprehensive Survey," *IEEE Access*, 2025.
[4] B. Kitchenham, O. P. Brereton, D. Budgen, M. Turner, J. Bailey, and S. Linkman, "Systematic literature reviews in software engineering – A systematic literature review," *Information and Software Technology*, vol. 51, no. 1, pp. 7–15, 2009.
[5] M. J. Page *et al.*, "The PRISMA 2020 statement: an updated guideline for reporting systematic reviews," *BMJ*, vol. 372, p. n71, 2021.
[6] M. Petticrew and H. Roberts, *Systematic Reviews in the Social Sciences: A Practical Guide*. Blackwell Publishing, 2006.
[7] Centre for Reviews and Dissemination, "Database of Abstracts of Reviews of Effects (DARE)," University of York, 2002.
[8] J. Cohen, "A coefficient of agreement for nominal scales," *Educational and Psychological Measurement*, vol. 20, no. 1, pp. 37–46, 1960.
[9] J. L. Fleiss, "Measuring nominal scale agreement among many raters," *Psychological Bulletin*, vol. 76, no. 5, pp. 378–382, 1971.
[10] J. D. de Arcaya, A. I. Torre-Bastida, G. Zárate, R. Miñón, and A. Almeida, "A Joint Study of the Challenges, Opportunities, and Roadmap of MLOps and AIOps," *ACM Computing Surveys*, 2023.
[11] L. Zhang, T. Jia, M. Jia, Y. Wu, A. Liu, Y. Yang, Z. Wu, X. Hu, P. Yu, and Y. Li, "A Survey of AIOps in the Era of Large Language Models," *ACM Computing Surveys*, 2025.
[12] A. Bandi, B. Kongari, R. Naguru, S. Pasnoor, and S. V. Vilipala, "The Rise of Agentic AI: A Review of Definitions, Frameworks, Architectures, Applications, Evaluation Metrics, and Challenges," *Future Internet*, 2025.
[13] M. A. Ferrag, N. Tihanyi, and M. Debbah, "From LLM Reasoning to Autonomous AI Agents: A Comprehensive Review," *IEEE Access*, 2026.
[14] Y. Hmimou, M. Tabaa, A. Khiat, and Z. Hidila, "A Multi-Agent System for Cybersecurity Threat Detection and Correlation Using Large Language Models," *IEEE Access*, 2025.
[15] N. Kshetri, "Transforming cybersecurity with agentic AI to combat emerging cyber threats," *Telecommunications Policy*, 2025.
[16] F. Piccialli, D. Chiaro, S. Sarwar, D. Cerciello, P. Qi, and V. Mele, "AgentAI: A comprehensive survey on autonomous agents in distributed AI for industry 4.0," *Expert Systems with Applications*, 2025.
[17] I. ADABARA, B. O. Sadiq, A. N. Shuaibu, Y. I. Danjuma, and M. Venkateswarlu, "A Review of Agentic AI in Cybersecurity: Cognitive Autonomy, Ethical Governance, and Quantum-Resilient Defense," *F1000Research*, 2025.
[18] I. Hasanov, S. Virtanen, A. Hakkala, and J. Isoaho, "Application of Large Language Models in Cybersecurity: A Systematic Literature Review," *IEEE Access*, 2024.
[19] F. Dehais, A. Lafont, R. N. Roy, and S. Fairclough, "A Neuroergonomics Approach to Mental Workload, Engagement and Human Performance," *Frontiers in Neuroscience*, 2020.
[20] S. Borsci, A. Malizia, M. Schmettow, F. van der Velde, G. Tariverdiyeva, D. Balaji, and A. Chamberlain, "The Chatbot Usability Scale: the Design and Pilot of a Usability Scale for Interaction with AI-Based Conversational Agents," *Personal and Ubiquitous Computing*, 2021.
[21] I. Albuquerque, A. Tiwari, M. Parent, R. Cassani, J.-F. Gagnon, D. Lafond, S. Tremblay, and T. H. Falk, "WAUC: A Multi-Modal Database for Mental Workload Assessment Under Physical Activity," *Frontiers in Neuroscience*, 2020.
[22] J. Mietkiewicz, A. N. Abbas, C. W. Amazu, G. Baldissone, A. Madsen, M. Demichela, and M. C. Leva, "Enhancing Control Room Operator Decision Making," *Processes*, 2024.
[23] W. Kasri, Y. Himeur, H. A. Alkhazaleh, S. Tarapiah, S. Atalla, W. Mansoor, and H. Al-Ahmad, "From Vulnerability to Defense: The Role of Large Language Models in Enhancing Cybersecurity," *Computation*, 2025.
[24] M. Uddin, M. Irshad, I. A. Kandhro, F. Alanazi, F. Ahmed, M. Maaz, S. Hussain, and S. S. Ullah, "Generative AI revolution in cybersecurity: a comprehensive review of threat intelligence and operations," *Artificial Intelligence Review*, 2025.
[25] I. ADABARA, B. O. Sadiq, A. N. Shuaibu, Y. I. Danjuma, and M. Venkateswarlu, "Trustworthy agentic AI systems: a cross-layer review of architectures, threat models, and governance strategies for real-world deployment," *F1000Research*, 2025.
[26] V. Chamola, V. Hassija, A. R. Sulthana, D. Ghosh, D. Dhingra, and B. Sikdar, "A Review of Trustworthy and Explainable Artificial Intelligence (XAI)," *IEEE Access*, 2023.
[27] F. Fu, H. Ding, Y. Qin, J. Yu, and D. Xu, "Leveraging multi-agent framework for root cause analysis," *Complex and Intelligent Systems*, 2026.
[28] X. Ji, L. Zhang, W. Zhang, F. Peng, Y. Mao, X. Liao, and K. Zhang, "LEMAD: LLM-Empowered Multi-Agent System for Anomaly Detection in Power Grid Services," *Electronics*, 2025.
[29] H. Zou, Y. Zhao, X. Chen, L. Wang, J. Yu, and L. Yuan, "Graph-Augmented Multi-Agent Robust Root Cause Analysis in AIOps," *Computers, Materials and Continua*, 2026.
[30] V. Avgerinos, K. Ramantas, L. Alonso, and C. Verikoukis, "ARM: Autonomous Remediation and Management With LLM Agents for Intent-Driven Control," *IEEE IoT Journal*, 2026.
[31] N. Fukuda, H. Nozue, and H. Oishi, "Small Language Model Agent for the Operations of Continuously Updating ICT Systems," *IEEE Access*, 2025.
[32] F. Y. Loumachi, M. C. Ghanem, and M. A. Ferrag, "Advancing Cyber Incident Timeline Analysis Through Retrieval-Augmented Generation and Large Language Models," *Computers*, 2025.
[33] I. Ismail, R. Kurnia, F. Widyatama, I. M. Wibawa, Z. A. Brata, Ukasyah, G. A. Nelistiani, and H. Kim, "Enhancing Security Operations Center: Wazuh Security Event Response with Retrieval-Augmented-Generation-Driven Copilot," *Sensors*, 2025.
[34] C.-H. Chou, A. Sudheer, and Y. Park, "An LLM-Based Agentic Network Traffic Incident-Report Approach Towards Explainable-AI Network Defense," *Journal of Sensor and Actuator Networks*, 2026.
[35] I. Ismail, R. Kurnia, Z. A. Brata, G. A. Nelistiani, S. Heo, H. Kim, H. Kim, H. Kim, and H. Kim, "Toward Robust Security Orchestration and Automated Response in Security Operations Centers with a Hyper-Automation Approach Using Agentic Artificial Intelligence," *Information*, 2025.
[36] G. Jones, D. Kasimatis, N. Pitropakis, R. Macfarlane, and W. J. Buchanan, "Analysing the role of LLMs in cybersecurity incident management," *International Journal of Information Security*, 2025.
[37] D. Toprani and V. K. Madisetti, "LLM Agentic Workflow for Automated Vulnerability Detection and Remediation in IaC," *IEEE Access*, 2025.

[38] W. Zhang, Z. Yang, F. Peng, L. Zhang, Y. Chen, and R. Chen, "GALR: Graph-Based Root Cause Localization and LLM-Assisted Recovery for Microservices," *Electronics*, 2026.
[39] A. Agrawal, M. Nadeem, A. A. Nuaim, and A. A. Nuaim, "Artificial intelligence driven multi agent framework for adaptive cyber attack simulation and automated incident response in cyber range environments," *Scientific Reports*, 2026.
[40] A. Rodríguez, J. Gómez, and A. Diaconescu, "A Decentralised Self-Healing Approach for Network Topology Maintenance," *Autonomous Agents and Multi-Agent Systems*, 2020.
[41] R. D. Zota, C. Bărbulescu, and R. Constantinescu, "A Practical Approach to Defining a Framework for Developing an Agentic AIOps System," *Electronics*, 2025.
[42] W. Jiang and F. Hu, "Artificial Intelligence Agent-Enabled Predictive Maintenance: Conceptual Proposal and Basic Framework," *Computers*, 2025.
[43] T. Ban, T. Takahashi, S. Ndichu, and D. Inoue, "Breaking Alert Fatigue: AI-Assisted SIEM Framework for Effective Incident Response," *Applied Sciences*, 2023.
[44] O. I. Falowo and J. B. Abdo, "Empirical Study on Automation, AI Trust, and Framework Readiness in Cybersecurity Incident Response," *Algorithms*, 2026.
[45] M. Landauer, S. Onder, F. Skopik, and M. Wurzenberger, "Deep learning for anomaly detection in log data: A survey," *Machine Learning with Applications*, 2023.
[46] S. Oesch, J. Hutchins, P. Austria, and A. Chaulagain, "Agentic AI and the Cyber Arms Race," *Computer*, 2025.
[47] M. A. Lopez, W. T. Lunardi, G. Lawton, and S. Thakkar, "Enhancing Autonomous System Security and Resilience With Generative AI: A Comprehensive Survey," *IEEE Access*, 2024.
[48] D. F. Pedroso, L. Almeida, L. E. G. Pulcinelli, W. A. A. Aisawa, I. Dutra, and S. M. Bruschi, "Anomaly Detection and Root Cause Analysis in Cloud-Native Environments Using Large Language Models and Bayesian Networks," *IEEE Access*, 2025.
[49] S. Kumar, S. Datta, V. Singh, D. Datta, S. K. Singh, and R. Sharma, "Applications, Challenges, and Future Directions of Human-in-the-Loop Learning," *IEEE Access*, 2024.

VALDOMIRO DE OLIVEIRA SOUZA JR. received the B.S. degree in information technology management from the Universidade do Vale do Rio dos Sinos (Unisinos), São Leopoldo, Brazil, in 2025, where he is currently pursuing the M.Sc. degree in applied computing with the Applied Computing Graduate Program (PPGCA).

From 1997 to 2012, he held infrastructure and operations positions in academic and industry organizations, including Pure Bros, a mobile value-added services integrator acquired by Zenvia in 2013. Since 2013, he has been with Zenvia, São Paulo, Brazil, where he has been Head of Information Technology since 2025. He leads the product, engineering, infrastructure, site reliability engineering (SRE), information security, observability, and FinOps teams of the company's communications platform as a service (CPaaS) business unit, whose messaging and voice platforms sustain peak loads above one million transactions per minute. He built Zenvia's SRE organization, which grew to 35 engineers across nine squads to support software as a service (SaaS), CPaaS, and corporate systems, led the improvements that supported the company's ISO/IEC 27001 certification in 2022, and was among the leaders of the technical integration of ten acquired companies. His research interests include agentic AI, AI-assisted incident response, the reduction of mean time to detect (MTTD) and mean time to recover (MTTR), and cognitive load in operations teams.

RAFAEL KUNST received the M.Sc. and Ph.D. degrees in computer science from the Federal University of Rio Grande do Sul (UFRGS), Porto Alegre, Brazil.

He is currently a Professor and a Researcher with the Applied Computing Graduate Program (PPGCA), Universidade do Vale do Rio dos Sinos (Unisinos), São Leopoldo, Brazil, where he currently serves as the Program Coordinator and is a member of the Software Innovation Laboratory (SOFTWARELAB). In 2020, he was a Visiting Professor with the IBM Watson IoT Center, Munich, Germany. He also collaborates with the Economics Graduate Program at Unisinos on data science topics and acts as an ad hoc Consultant for the Brazilian Ministry of Education. His research interests include next-generation mobile communications (5G and 6G), Industry 4.0, smart cities, the Internet of Things, and the application of big data analytics and machine learning to telecommunications.